THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE TWO-CHAPTER TEST SET D

Chapters 9 & 10: **The Amazing World of Solutes, Solvents and Solutions, Light: Mirrors and Lenses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An object floats in water if its density is: 1

(a) more than water **(b)** infinite **(c)** equal to zero **(d)** less than water

2. The floating of ice is important because: 1

(a) it insulates water below for aquatic life
(b) it makes water salty
(c) it removes oxygen
(d) it heats the lake

3. To make an egg float, we can: 1

(a) add salt to the water **(b)** add more tap water
(c) remove the egg shell **(d)** warm the water

4. Brass can be called a solid–solid solution because: 1

(a) it is a compound **(b)** it is a liquid
(c) it is a gas **(d)** zinc is mixed uniformly in copper

5. The mass of 1 cm^3 of water is about: 1

(a) 10 g **(b)** 100 g **(c)** 1 g **(d)** 0.1 g

6. Solar cookers use ___ mirrors to focus sunlight. 1

(a) convex **(b)** plane **(c)** concave **(d)** no

7. A concave lens makes light rays: 1

(a) stop **(b)** diverge **(c)** stay parallel **(d)** converge

8. A pencil appears bent in a glass of water because of: 1

(a) magnetism **(b)** reflection from water
(c) absorption **(d)** refraction of light

9. Which pair is correctly matched? 1

(a) Convex lens – converging **(b)** Convex mirror – converging
(c) Concave lens – converging **(d)** Plane mirror – diverging

10. The Moon is visible because it: 1

(a) absorbs all sunlight **(b)** produces its own light
(c) refracts only **(d)** reflects sunlight

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Why does soda lose its fizz faster when warm? 2

12. What is the role of a weighing balance and a measuring cylinder? 2

13. Why is the Moon seen in the sky although it has no light of its own? 2

14. Name the two types of reflection by smooth and rough surfaces. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Explain how density helps identify substances. 3

16. Explain how the tilt of a mirror changes the angle of incidence for the same ray. 3

17. Explain why a mixture of two liquids may be called a solution. 3

OR

What would happen to a saturated solution on cooling and why? 3

18. Explain the term ‘field of view’ and which mirror gives the largest. 3

OR

List three differences between a concave mirror and a concave lens. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Describe how a solute behaves when a saturated solution is cooled and show it with a crystallisation example. 5

OR

Write a note on the importance of solutions in nature and industry. 5

20. Describe the activity to compare the images of text seen through a convex lens, concave lens and a flat glass. 5

OR

Describe diffuse reflection and regular reflection with examples and explain why we can see non-luminous objects. 5

END OF QUESTION PAPER — Total: 40 Marks

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